

Problem Code: SIH26202 (Smart Automation)	Target Agency: MeitY / NIC / Welfare Ministries	Design Standard: GoI UX4G 3.0 Standard
Officer Demo: officer@welfare.gov.in (Officer@2026)	Admin Demo: admin@welfare.gov.in (Admin@2026)	Citizen Demo: Public OTP Login (123456)

1. Executive Problem & Core Solution Architecture

The Problem: India disburses over ₹52 lakh crore annually via Direct Benefit Transfer (DBT). A critical fraction is lost to leakage: ghost recipients, deceased claimants, duplicate Aadhaar numbers, and ineligible tax-payers pocketing BPL assistance. Manual inspections cannot scale across hundreds of millions of records, and legacy systems act as 'black boxes'—alerting officers without explainability, offering no case workflows, and giving citizens no recourse.

The Closed-Loop Triad: This prototype demonstrates an interconnected architecture: **ML Scoring Engine (Isolation Forest)** → **Welfare Officer Case Management (Kanban & SLA)** → **Immutable Audit Trail (RS256 RBAC)** → **Citizen Feedback Channel (Grievance Portal)**.

1. Explainable AI (XAI) Decomposes anomaly scores into exact feature contribution percentages (e.g., 34% income mismatch, 21% geo-anomaly) instead of opaque black-box alerts.	2. Privacy by Design Full 12-digit Aadhaar numbers are never stored or transmitted. The API enforces strict last-4 digit masking (XXXX-XXXX-1234) adhering to UIDAI regulations.
3. Citizen-in-the-Loop Empowers citizens to file grievances with last-4 Aadhaar and OTP verification, turning ground-level complaints into real-time verification signals.	4. Actionable Kanban & SLAs Transforms static data into operational case management: drag-and-drop triage (Under Review → Confirmed → Resolved) with live SLA countdown chips.
5. Multi-Profile RBAC Role-based access separating Field Officers, System Administrators, and Public Citizens with asymmetric JWT (RS256) and token rotation.	6. Immutable Audit Trail Every officer review, flag transition, search query, and report export is immutably logged with actor ID, IP address, and timestamp for total governance.

2. Complete Video Walkthrough Script (Visual Action vs. Voiceover)

Follow the timeline below while recording your screen. Keep cursor movements deliberate, highlight key data points, and deliver narration at a calm, authoritative pace.

SECTION 1: Hook & The National Problem Statement (0:00 – 0:45)

Timestamp	Visual Action on Screen (What to Show & Click)	Voiceover Narration (Spoken Script)
0:00 – 0:20	- Show login screen at <code>http://localhost:5173</code>. - Quick glance of UI elements: Indian emblem, UX4G styling, role tabs (Officer, Admin, Citizen).	"India disburses over ₹52 lakh crore annually through Direct Benefit Transfer schemes like PM-KISAN, Ayushman Bharat, and NSAP pensions. Yet, significant funds are drained through leakage: ghost beneficiaries, duplicate accounts, dead persons drawing pensions, and wealthy tax-payers claiming welfare meant for the poorest families."
0:20 – 0:45	- Show high-level system flow: ML Scoring → Officer Review → Audit Trail → Citizen Feedback. - Highlight the closed-loop nature of the platform.	"Manual scrutiny cannot scale to 1.4 billion citizens. Legacy alert tools fail because they act as black boxes—giving officers no explainability, no case workflow, and leaving citizens with no recourse. Today, we present our AI Welfare Leakage Detection System —an end-to-end platform bridging Citizens, Welfare Officers, and Administrators with explainable AI, privacy-by-design, and actionable workflows."

SECTION 2: Profile 1 — The Citizen Portal (Public Recourse) (0:45 – 1:40)

Key USPs: *Privacy by Design (Last-4 Aadhaar Only) + Ground-Truth Grievance Feedback Loop*

Timestamp	Visual Action on Screen (What to Show & Click)	Voiceover Narration (Spoken Script)
0:45 – 1:00	- On Login page, click 'Citizen' tab. - Click 'Continue as Citizen' (no password needed). - Land on Citizen Grievance Portal (/citizen).	"We begin with our first profile: the Citizen . Citizens do not need complicated accounts. They serve as our ground-truth verification layer, reporting discrepancies directly into our detection pipeline."
1:00 – 1:20	- Click 'Submit Grievance'. - Point out 4-step progress stepper: <i>Details</i> → <i>Documents</i> → <i>OTP Verify</i> → <i>Confirm</i>. - Type Aadhaar Last 4: 4821. - Select Scheme (<i>PM-KISAN</i>), select Issue (<i>Payment Not Received</i>), enter description, upload a mock document.	"Notice our primary USP: Privacy by Design . In compliance with UIDAI regulations, full 12-digit Aadhaar numbers are never requested or stored— only the last 4 digits . The citizen completes an intuitive 4-step wizard: picking their scheme, categorizing their issue, providing details, and attaching supporting documents with client-side validation."
1:20 – 1:35	- Click 'Next', click 'Send OTP'. - Notice the 30s resend cooldown. - Enter demo OTP 123456 and submit. - Show generated Tracking ID: GRV-2026-48291.	"Following secure OTP verification, an encrypted tracking ID is instantly generated. This grievance immediately feeds into the officer detection pipeline as ground-truth evidence."
1:35 – 1:50	- Click 'Track Grievance' (/citizen/track). - Enter Tracking ID GRV-2026-48291 and Aadhaar Last-4 4821. - Click Search to reveal the vertical timeline.	"Citizens can track their case at any moment through a live vertical timeline—from Submitted, to Under Review, to Resolved—complete with resolution notes and SLA transparency."

SECTION 3: Profile 2 — Welfare Officer Portal (Operations & Case Review) (1:50 – 3:20)

Key USPs: *Explainable AI (XAI) + Asymmetric UX4G Dashboard + Kanban Triage with SLA Timers*

Timestamp	Visual Action on Screen (What to Show & Click)	Voiceover Narration (Spoken Script)
1:50 – 2:05	- Return to Login, select 'Officer'. - Sign in with officer@welfare.gov.in / Officer@2026. - Land on Detection Dashboard (/officer).	"Now, let's switch to our second profile: the Welfare Officer , protected by RS256 JWT authentication and strict server-side RBAC. The officer arrives at the Detection Dashboard , designed under the Government of India UX4G design standard."
2:05 – 2:25	- Point out asymmetric KPI grid: Flagged Today (hero card with sparkline), Resolved This Week, Pending Grievances, Registered Beneficiaries. - Highlight live count-up numbers and recent anomaly flags table with mini scorebars.	"Rather than generic equal cards, our dashboard uses an asymmetric layout prioritizing operational urgency. The hero card spotlights 'Flagged Today' with directional deltas and animated sparklines. Below, live anomaly feeds display mini scorebars and masked Aadhaar tags for immediate assessment."
2:25 – 2:45	- Click 'Beneficiaries' in sidebar. - Filter by State (<i>Maharashtra</i>), Scheme (<i>PM-KISAN</i>), and Flag Type (<i>Income Mismatch</i>). - Demonstrate instant server-side filtered search.	"Under Beneficiaries , officers can perform multi-criteria searches across 500+ records—filtering by state, scheme, or anomaly pattern, and sorting by risk score."
2:45 – 3:15	- Click on Ramesh Kumar Sharma to open Beneficiary Detail (/officer/beneficiaries/b1). - Highlight the Radial Anomaly Gauge (92.4 High). - Linger over Explainable Flag Reasons progress bars. - Scroll to DBT Transaction History & SLA chips.	"This is the centerpiece of our prototype— Explainable AI (XAI) . Instead of an unhelpful black-box alert, our system breaks down the 92.4 score into exact feature contributions: 34% from Income Mismatch : The beneficiary reports ₹12 lakh annual income while claiming low-income benefits. 21% from Geolocation Anomaly : Logins detected from Karnataka against a Maharashtra address. Officers also inspect the full DBT disbursement ledger with UTR numbers and active flags backed by live SLA chips."
3:15 – 3:35	- Click 'Flag Queue' (/officer/flags). - Drag a card from <i>Under Review</i> to <i>Confirmed</i>. - Check two cards and click 'Resolve All'. - Point out the countdown SLA chips.	"To make detection operational, the Flag Queue provides an agile Kanban workflow. Officers drag and drop cases between Under Review, Confirmed, and Resolved, trigger bulk actions, and monitor countdown timers before SLAs breach."

SECTION 4: Advanced Analytics & Profile 3 — System Administrator (3:35 – 4:25)

Key USPs: *Anomaly Heatmap + Transparent ML Metrics + Tamper-Proof Audit Logging*

Timestamp	Visual Action on Screen (What to Show & Click)	Voiceover Narration (Spoken Script)
3:35 – 3:55	- Click 'Heatmap' (/officer/heatmap). - Toggle metrics: <i>Leak Rate</i> → <i>Flag Count</i> → <i>Avg Score</i>. - Point out state criticality color codes.	"In the Anomaly Heatmap , spatial analytics identify macro leakage clusters across states like UP, Maharashtra, and Bihar. Administrators can toggle between Leak Rates, Flag Counts, or Average Scores to allocate ground-level investigative squads."
3:55 – 4:10	- Click 'ML Insights' (/officer/ml-insights). - Showcase metric tiles: Precision (91%), Recall (87%), F1 (89%), AUC-ROC (94%). - Show feature importance ranking chart.	"In ML Insights , we maintain total algorithmic transparency. We present the exact evaluation metrics of our Isolation Forest model—achieving an AUC-ROC of 94%—alongside relative feature importance weights."
4:10 – 4:30	- Logout and log in as Admin (admin@welfare.gov.in / Admin@2026). - Note title switches to 'System Overview' with purple ADMIN pill. - Show admin KPIs: Active Officers (34), Audit Events (1,284). - Click 'Audit Log' in sidebar.	"Finally, we introduce our third profile: the System Administrator . The interface dynamically transforms: the header displays the Admin pill, system-wide management KPIs appear, and the restricted Audit Log unlocks. Every officer action—beneficiary lookups, status transitions, and data exports—is recorded in an append-only audit trail with actor IDs, IP addresses, and timestamps for absolute accountability."

SECTION 5: Summary of USPs & Closing Impact (4:30 – 5:00)

Timestamp	Visual Action on Screen (What to Show & Click)	Voiceover Narration (Spoken Script)
4:30 – 4:50	- Navigate to Reports (/officer/reports). - Select date range, click 'Export' (CSV downloads). - Return to Detection Dashboard.	"To summarize, our solution delivers 5 decisive advantages: 1. Explainable AI that translates statistical anomalies into human-readable reasons. 2. Citizen-in-the-Loop validation that captures ground truth. 3. Privacy by Design with zero raw Aadhaar exposure. 4. Actionable Kanban Operations with strict SLA compliance. 5. Institutional Security with RS256 JWT rotation, Zod validation, and immutable auditing."
4:50 – 5:00	- Show title/closing slide with team credentials and contact details. - Fade to black.	"By stopping welfare leakage before money is disbursed, our prototype ensures that public welfare funds reach the genuine citizens who need them most. Thank you!"

3. Practical Recording Tips & Demo Checklist

- 1. Display & Resolution:** Record your browser in Fullscreen (F11) at 1920×1080 resolution with 100% or 110% zoom so that JetBrains Mono figures, SLA chips, and sparklines are crisp.
- 2. Intentional Cursor Flow:** When explaining features (e.g. the 92.4 score or 34% income mismatch), move the pointer smoothly to the element and hold for 0.5s before moving on.
- 3. Emphasize Key Nuances:** Judges specifically look for Explainability (why a flag was raised) and Privacy (Aadhaar masking). Call both out clearly during sections 2 and 3.
- 4. Audio Setting:** Use a dedicated headset or USB mic. Keep background noise to zero and speak at an energetic, conversational pace (approx. 130–140 words per minute).